

A Short Practical Introduction to Bayesian Analysis

Opher Donchin

24–25 September 2026

A short worked example

The Bayesian toolkit

The Bayesian workflow

Working through the workflow: golf putting

Becoming independent Bayesian analysts

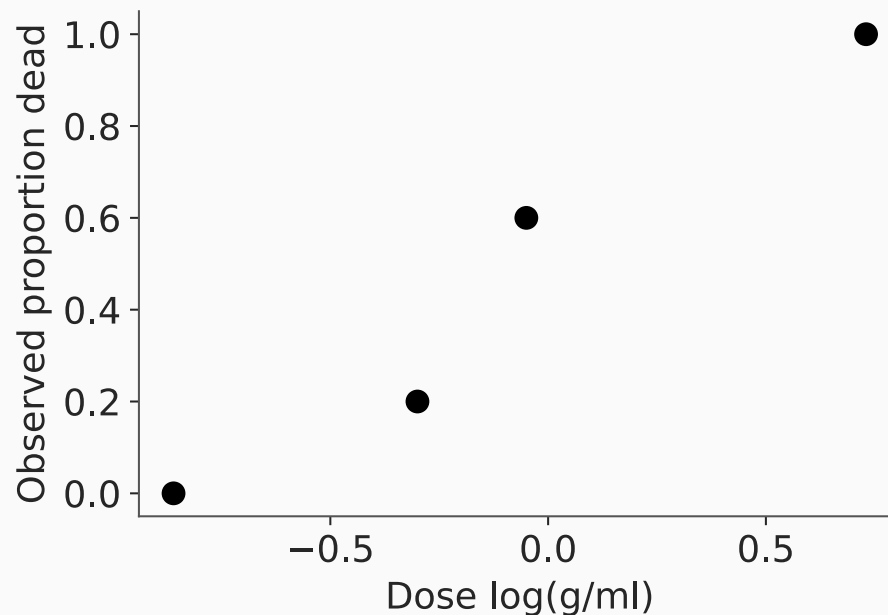
A short worked example

Dose response

Four doses, twenty animals

A toxin is given to rats at four doses, five rats per dose.

log dose	rats	deaths
-0.86	5	0
-0.30	5	1
-0.05	5	3
0.73	5	5

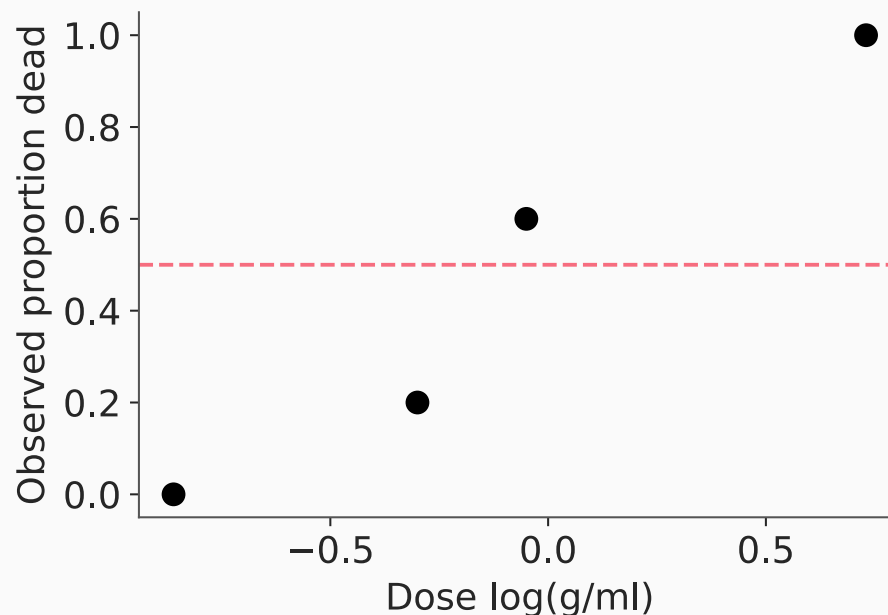


Data: Racine-Poon et al. (1986); Gelman & Vehtari, *Bayesian Workflow*, §3.5

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A generative model for mortality

$$y_j \sim \text{Binomial}(n_j, p_j)$$

$$\text{logit}(p_j) = \alpha + \beta x_j$$

$$\alpha \sim \text{Normal}(0, 5)$$

$$\beta \sim \text{HalfNormal}(5)$$

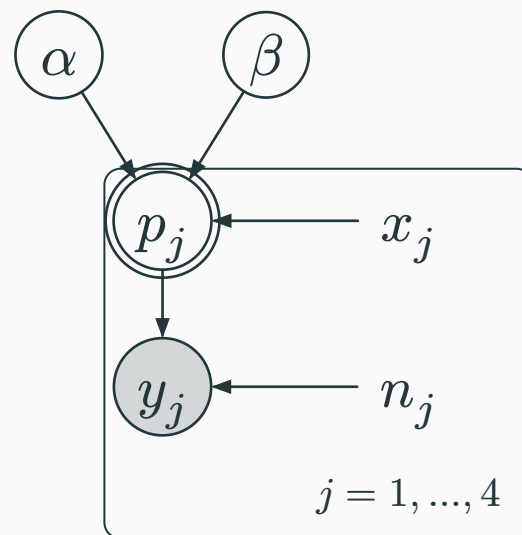
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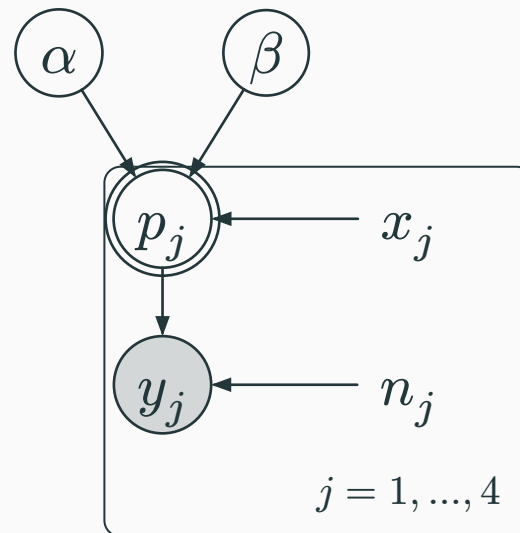
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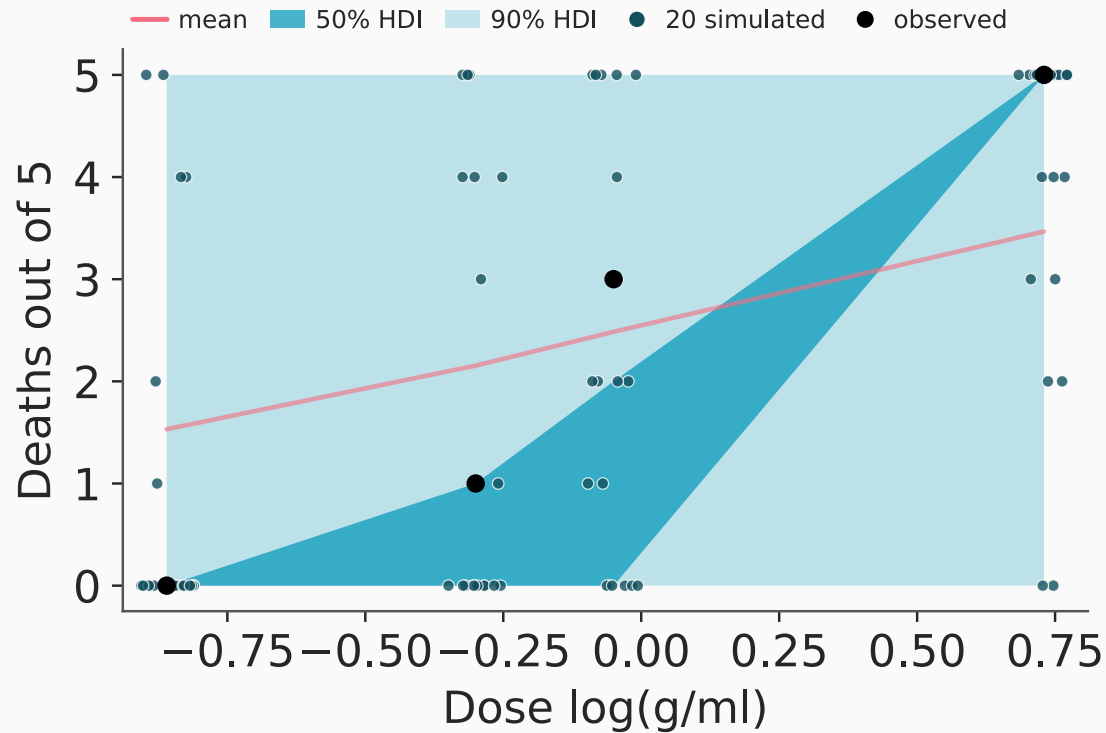
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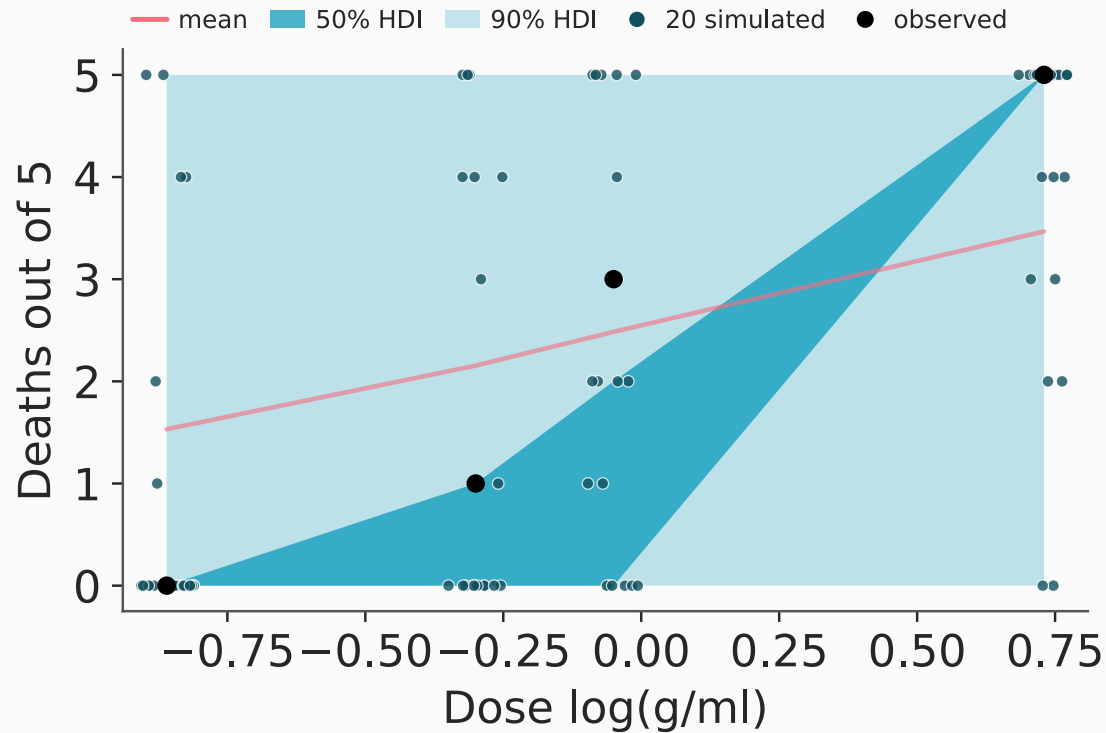
```
with pm.Model() as model:
    alpha = pm.Normal("alpha", mu=0, sigma=5)
    beta = pm.HalfNormal("beta", sigma=5)
    logit_p = alpha + beta * dose
    pm.Binomial("deaths", n=n, logit_p=logit_p, observed=deaths)
```

Priors: what could we have seen?



Prior predictive data should not

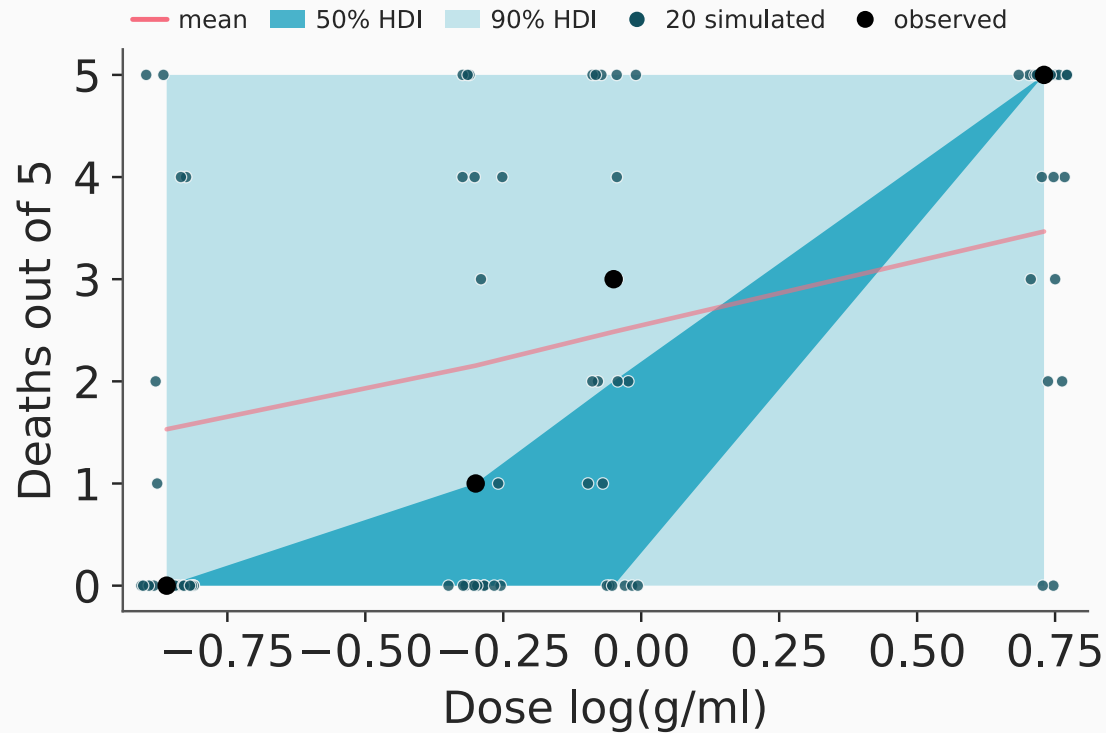
Priors: what could we have seen?



Prior predictive data should not

- violate common sense

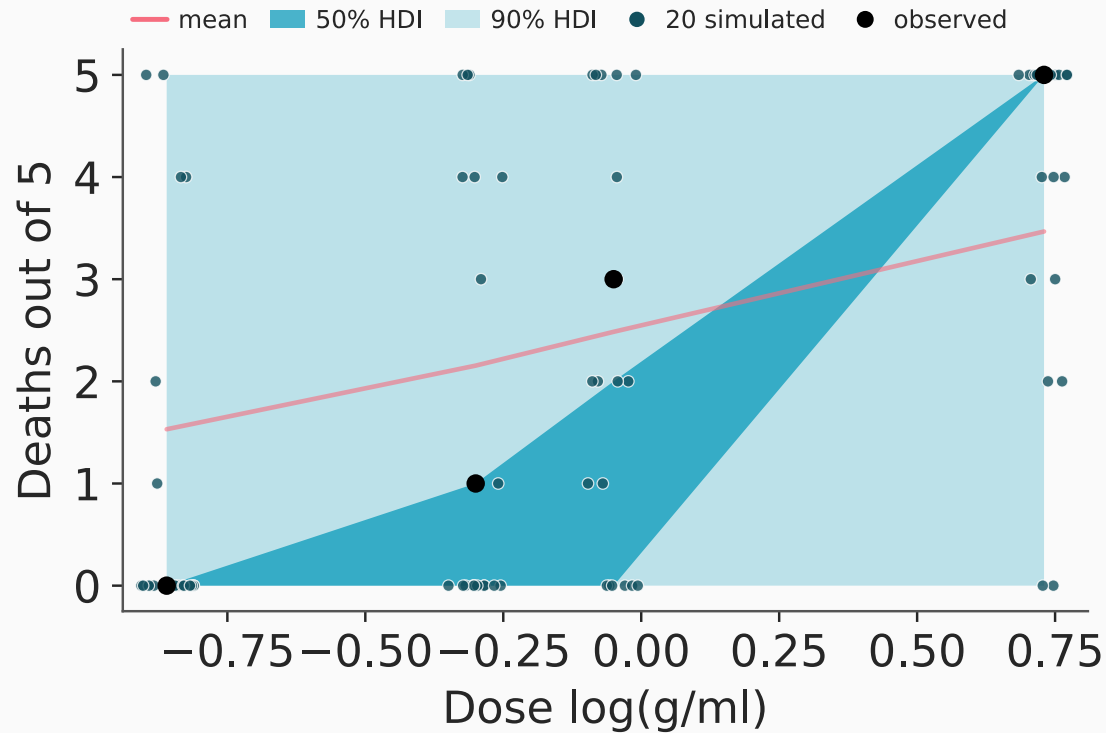
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Priors: what could we have seen?



Prior predictive data should not

- violate common sense
- make our data highly unlikely
- confine the model to our data

Sampling and diagnostics

Fitting the model means drawing samples from the posterior distribution.

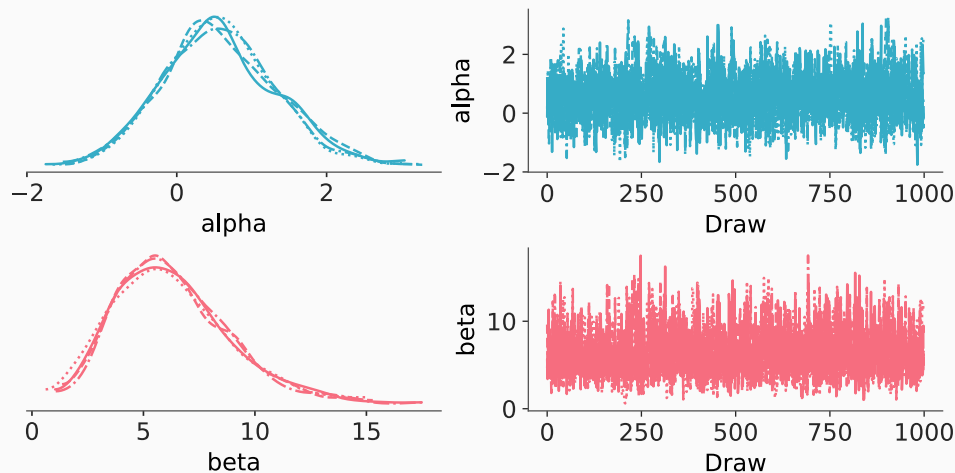
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    idata = pm.sample(draws=1000, chains=4)
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Sampling and diagnostics

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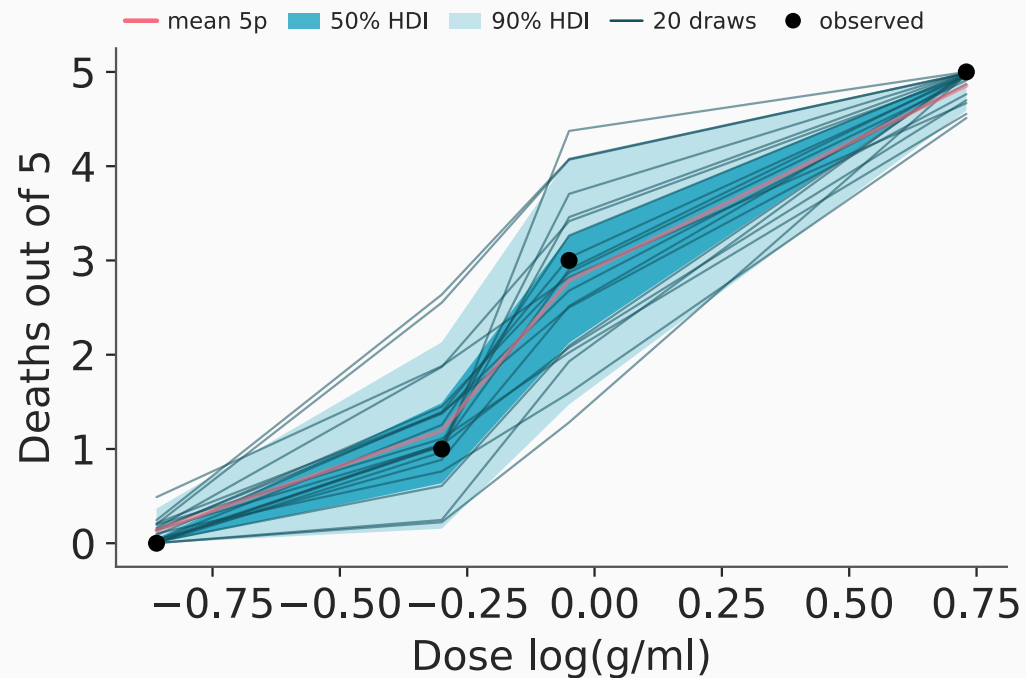


Check before interpreting:

- 0 divergences
- $\hat{R} = 1.00$
- ESS > 1,300 of 4,000 draws

Posterior: which dose-response curves remain plausible?

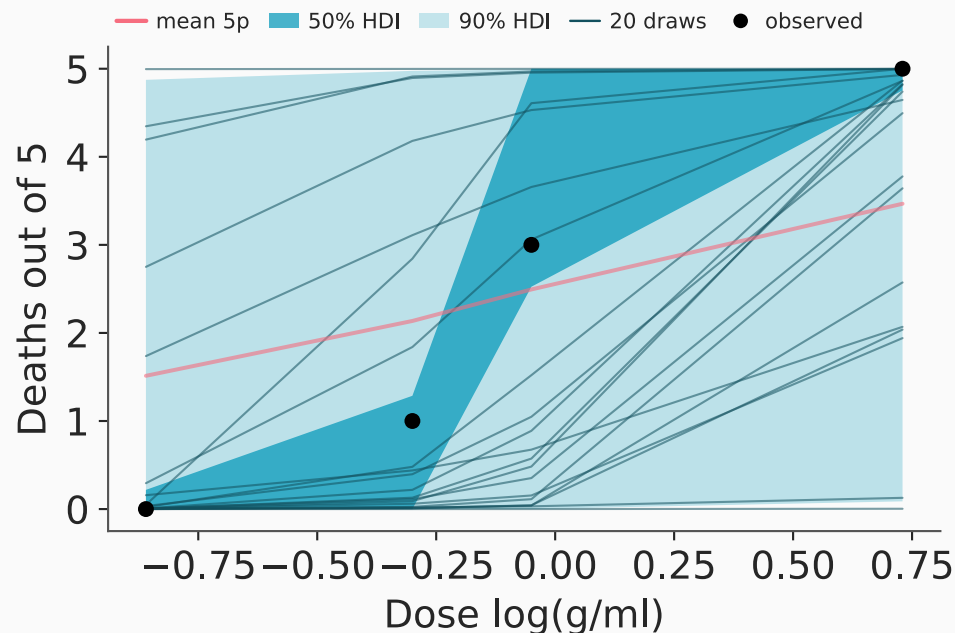
The bands show uncertainty about the mean number of deaths.



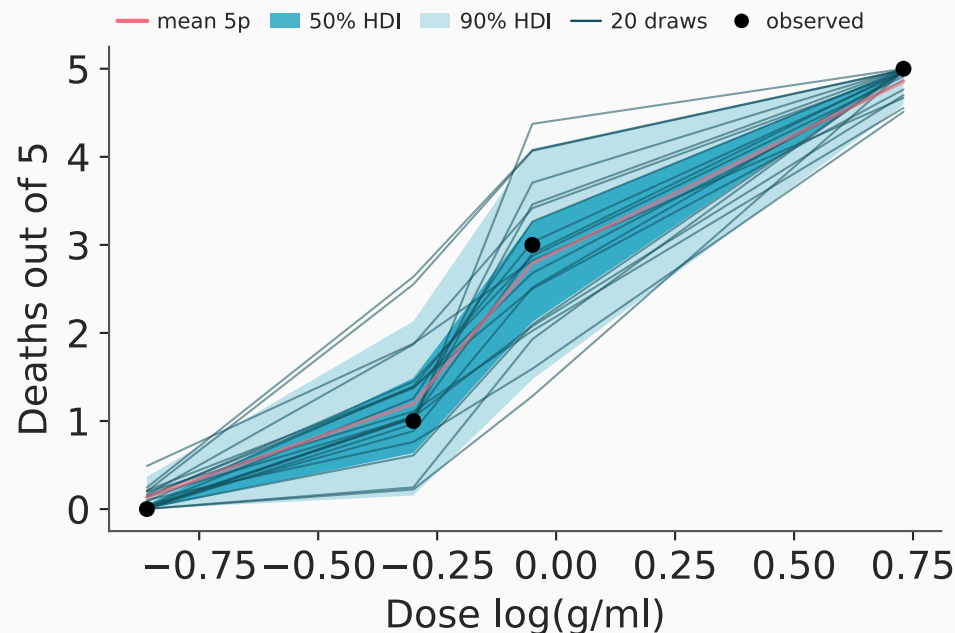
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Prior

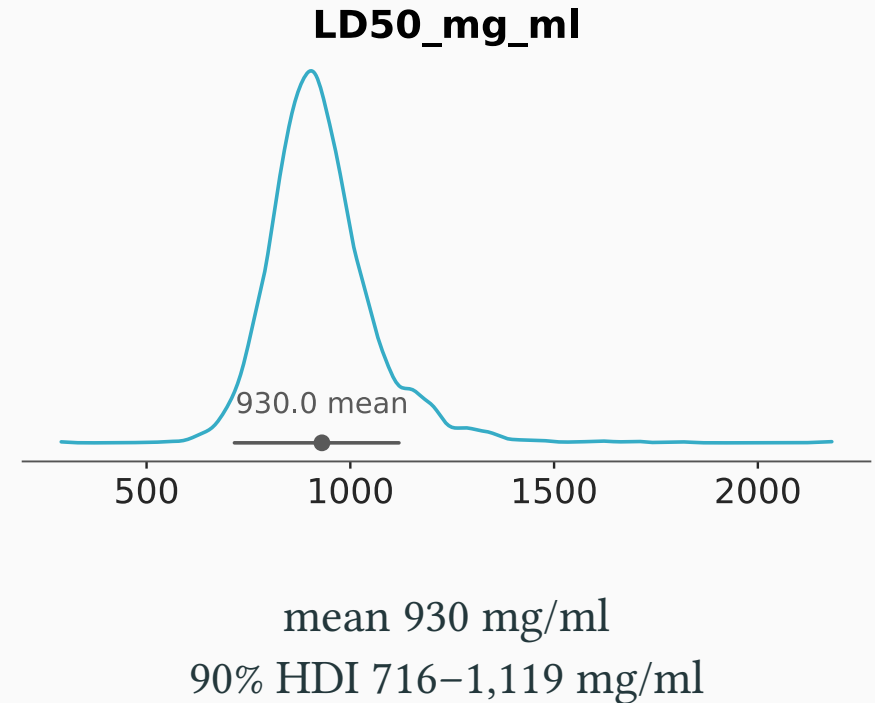


Posterior



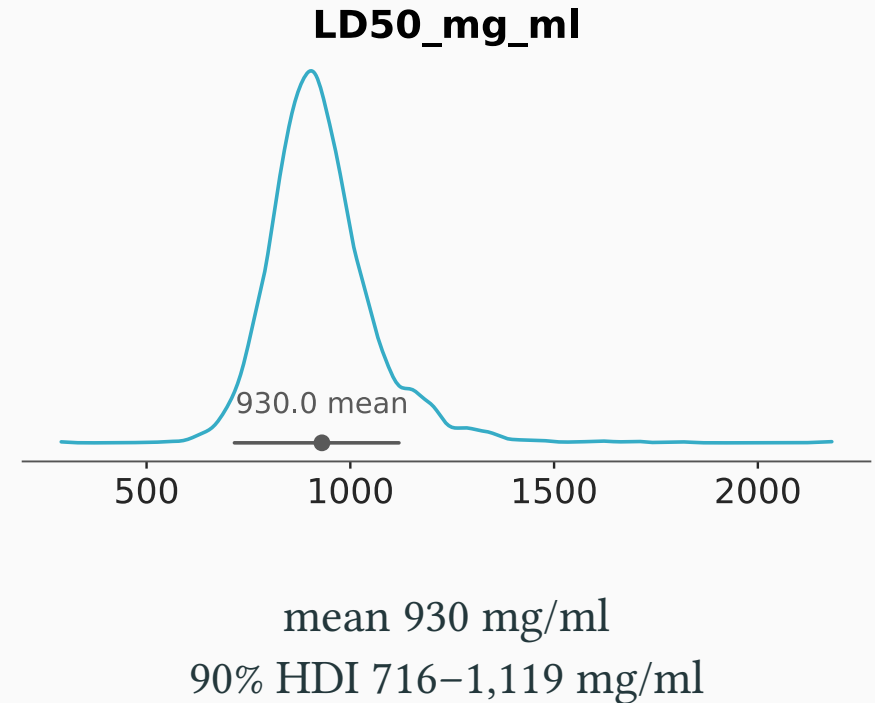
LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.



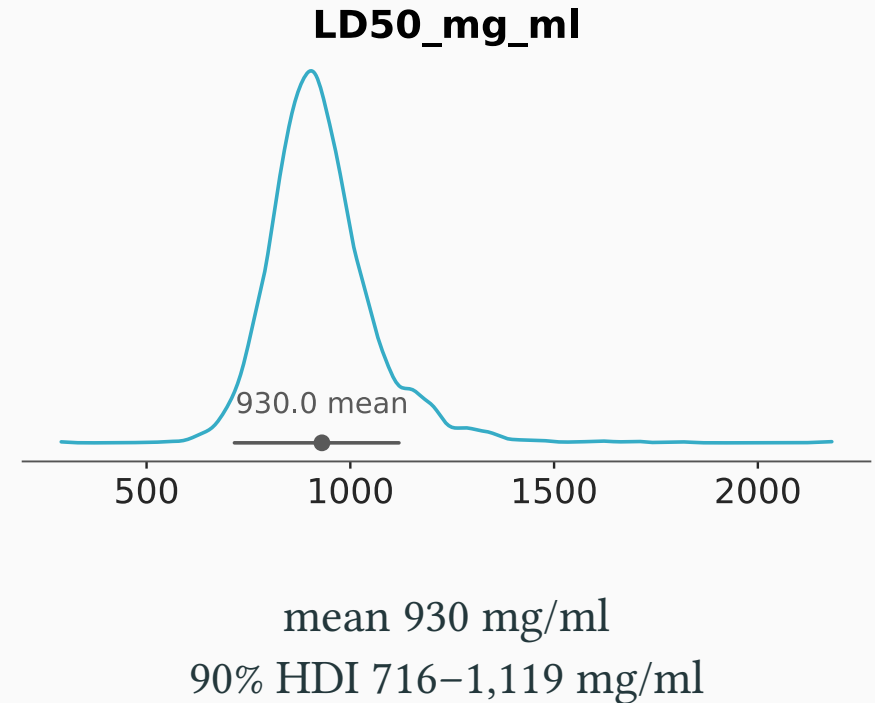
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- So every quantity comes with its own uncertainty.

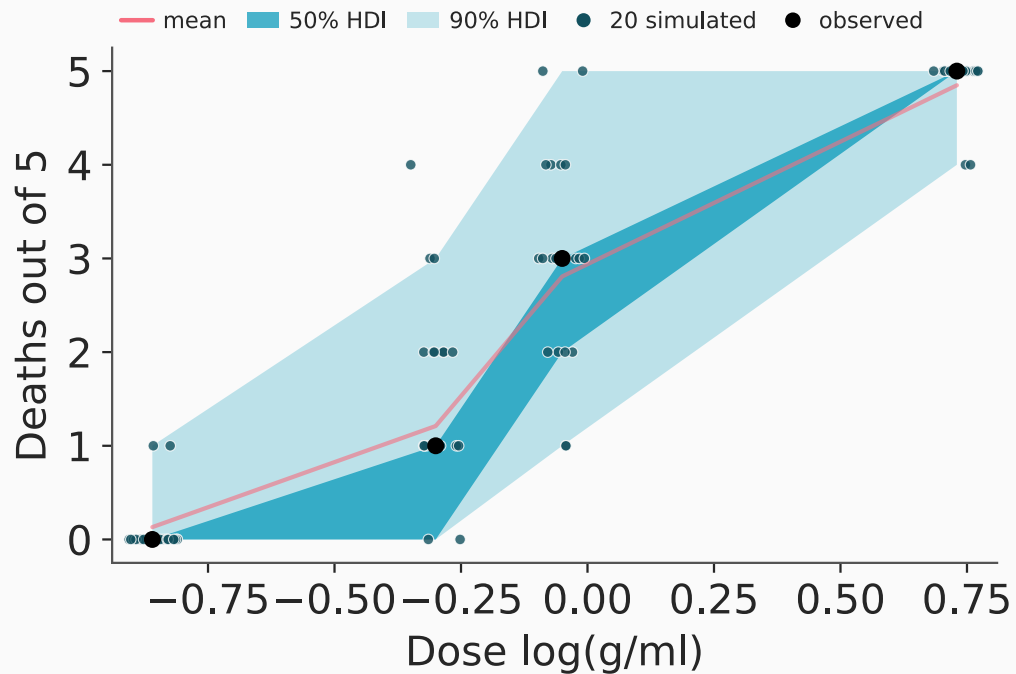


LD50: the quantity we actually care about

- Any quantity can be computed from the parameters:
 - $LD50 = -\alpha/\beta$
 - Once for every posterior draw.
- So every quantity comes with its own uncertainty.
- Here *how well* we know LD50 matters as much as its value.

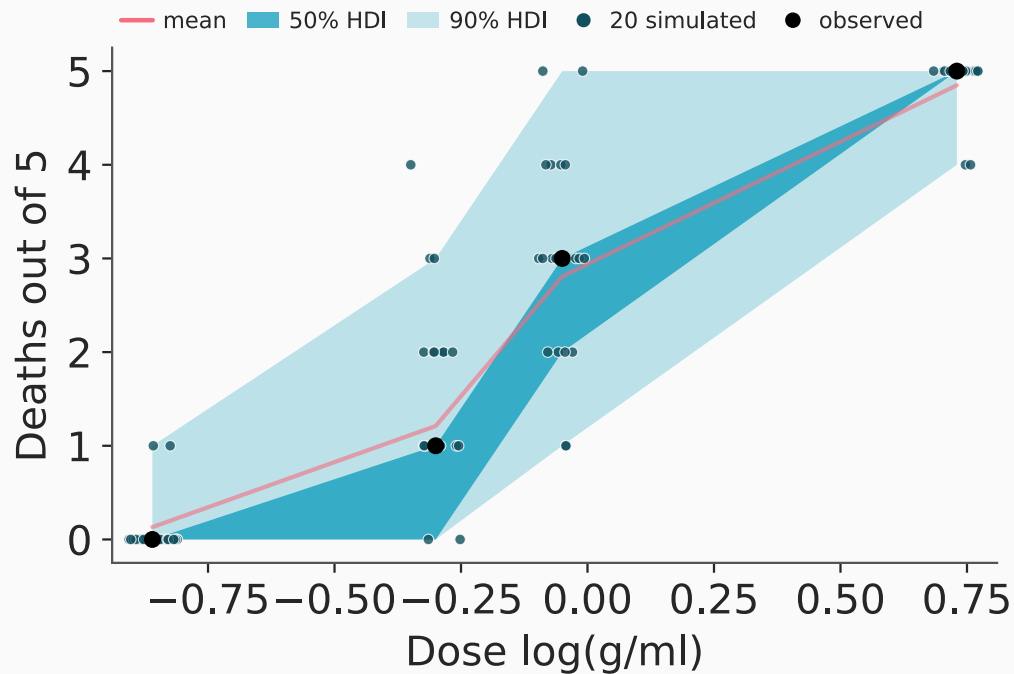


Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

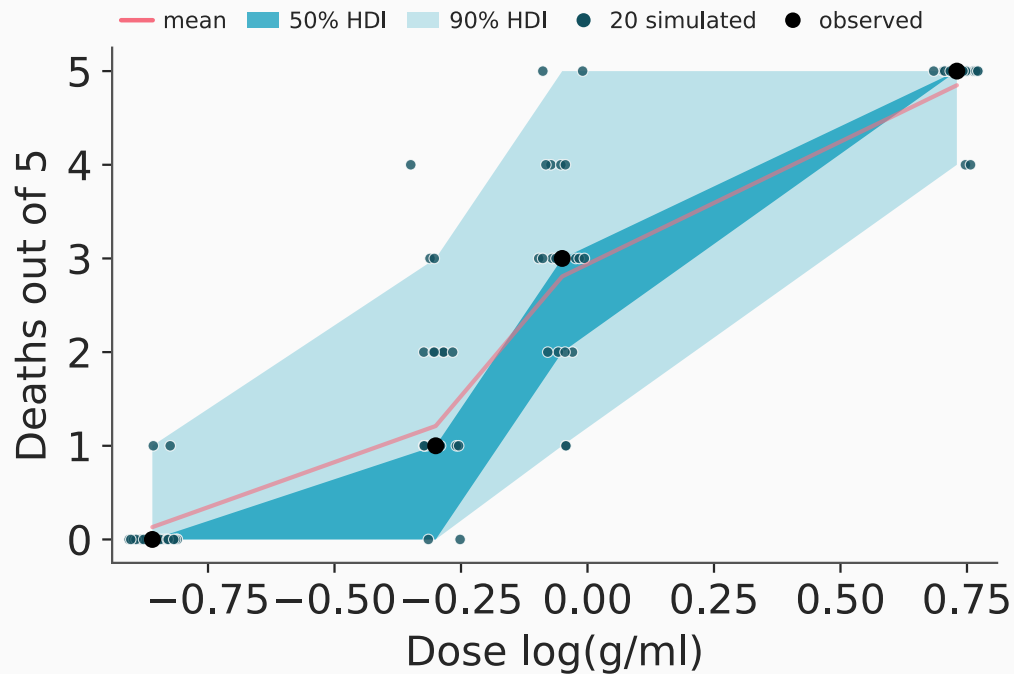
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- A workflow check meant to be done before looking at the posterior.

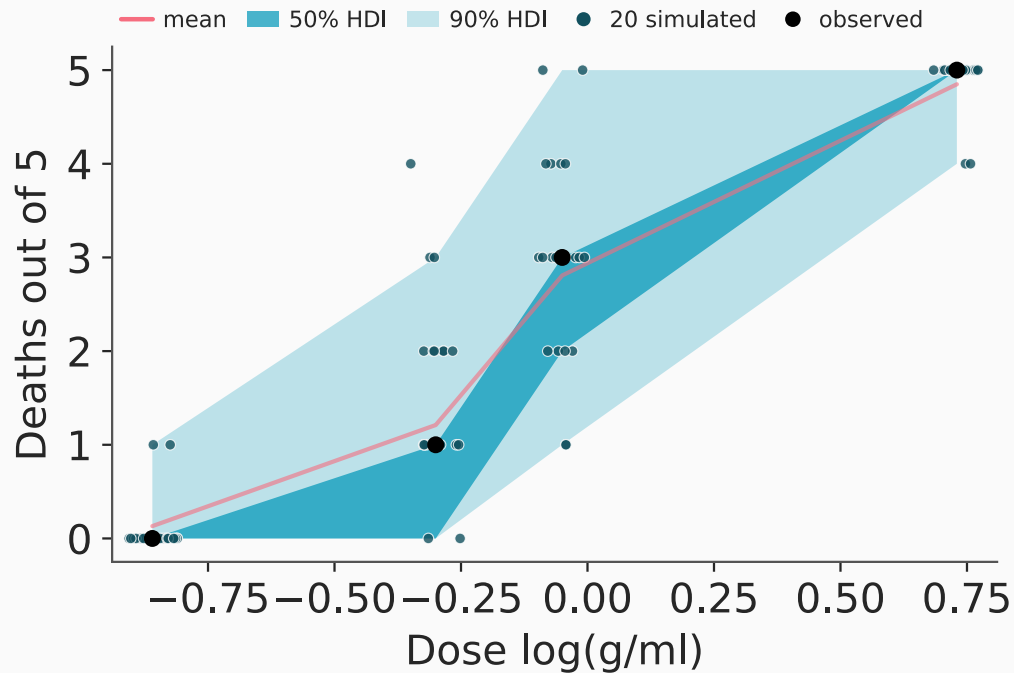
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- Workflow is not lockstep: going back to check is always fine.

Posterior predictive: can the model reproduce the experiment?



This is what the model predicts we could have seen, *after* fitting.

- A workflow check meant to be done before looking at the posterior.
- Workflow is not lockstep: going back to check is always fine.
- We understand data better than we understand parameters.

The Bayesian toolkit

Four libraries, one workflow



Write the model, sample the posterior

pymc.io



Choose and check priors

preliz.readthedocs.io



Diagnose, summarize, and plot

python.arviz.org



Regression models from a formula

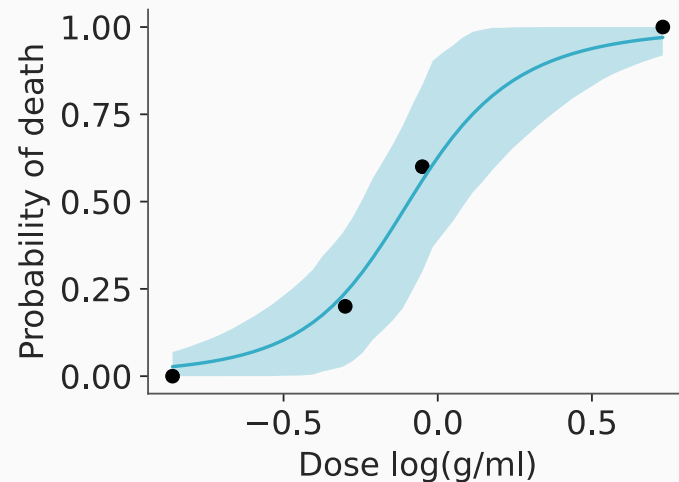
bambinos.github.io/bambi

Bambi: the same model as one formula

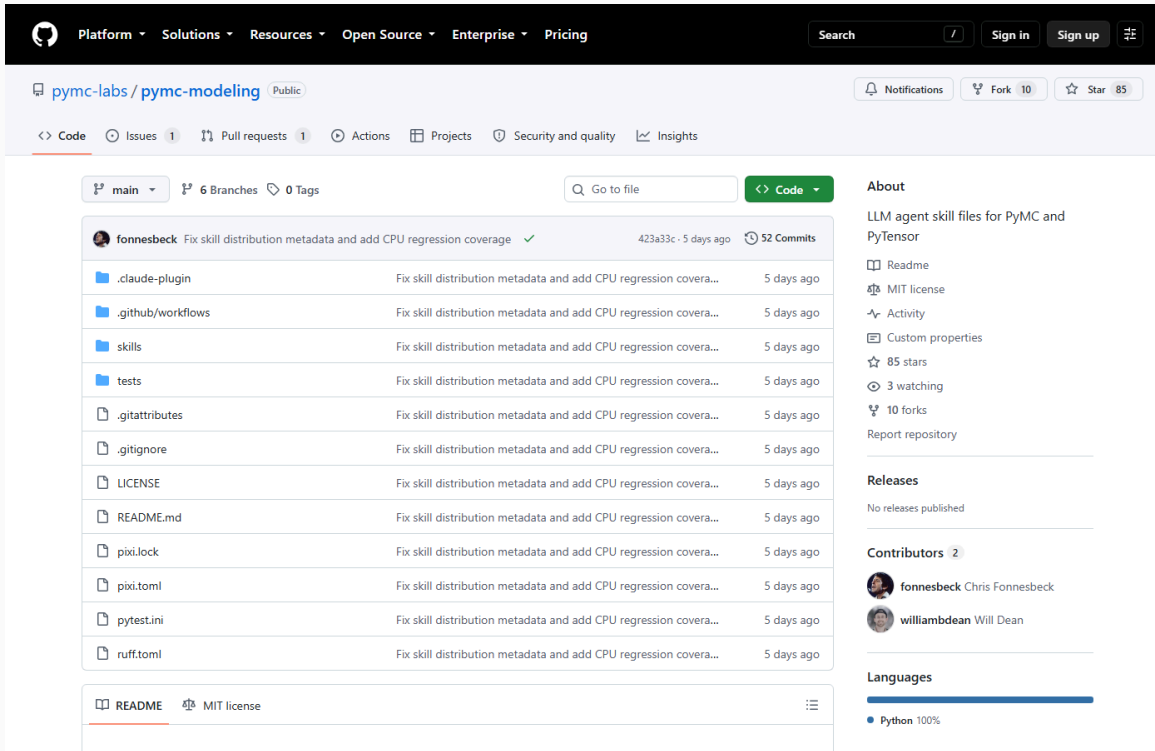
```
priors = {  
    "Intercept": bmb.Prior("Normal", mu=0, sigma=5),  
    "dose": bmb.Prior("HalfNormal", sigma=5),  
}  
model = bmb.Model("p(deaths, n) ~ dose", data,  
                  family="binomial", priors=priors)  
idata = model.fit()  
bmb.interpret.plot_predictions(model, idata, "dose")
```

Bambi: the same model as one formula

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Coding assistants can learn current PyMC practice

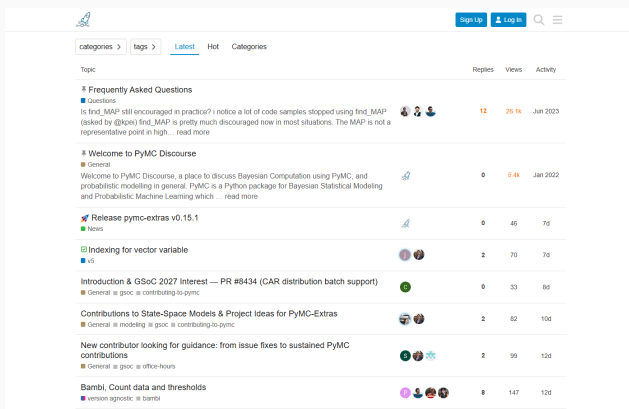


github.com/pymc-labs/pymc-modeling

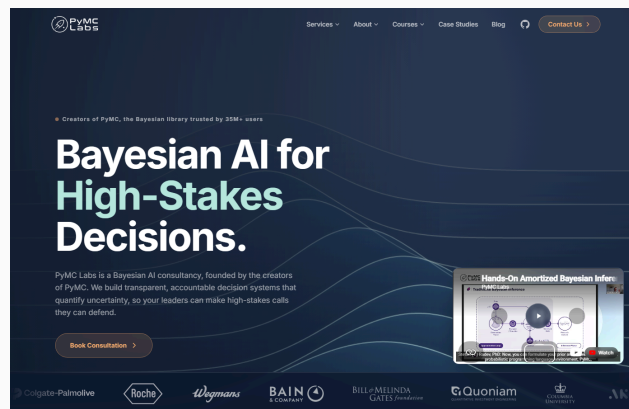
Skill files from PyMC Labs that teach AI coding assistants current PyMC and ArviZ practice.

We used them to build this course's notebooks.

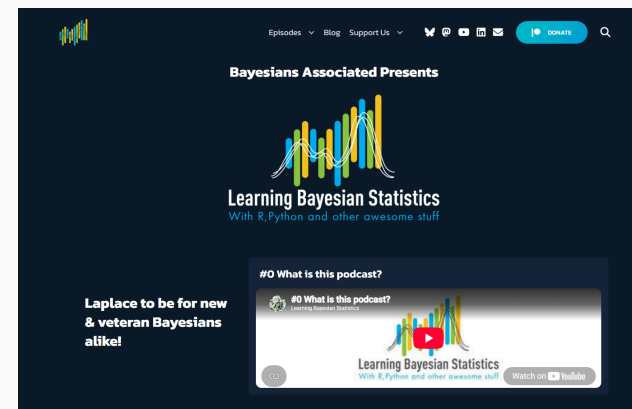
Where to ask, hire, and listen



Ask: PyMC Discourse
discourse.pymc.io



Hire: PyMC Labs
pymc-labs.com



Listen: Learning Bayesian Statistics
learnbayesstats.com

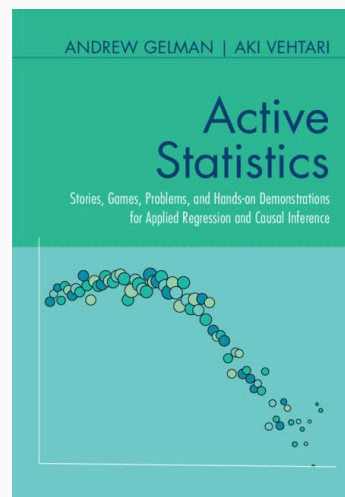
Books worth reading next



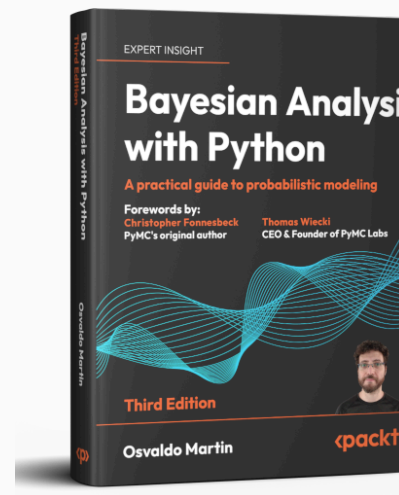
McElreath
Statistical Rethinking



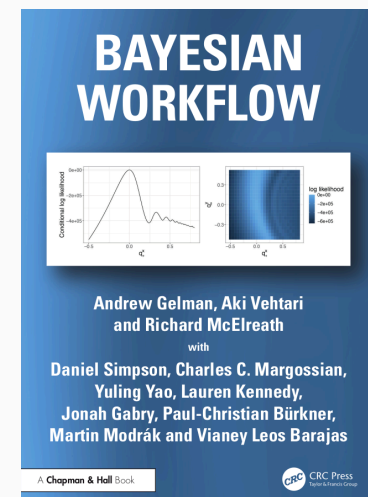
Gelman, Hill & Vehtari
Regression and Other Stories



Gelman & Vehtari
Active Statistics



Martin
Bayesian Analysis with Python



Gelman, Vehtari et al.
Bayesian Workflow

The Bayesian workflow

The workflow starts with a four-step loop



Data

The workflow starts with a four-step loop



The workflow starts with a four-step loop



The workflow starts with a four-step loop



The workflow starts with a four-step loop



Even at this level, we allow for

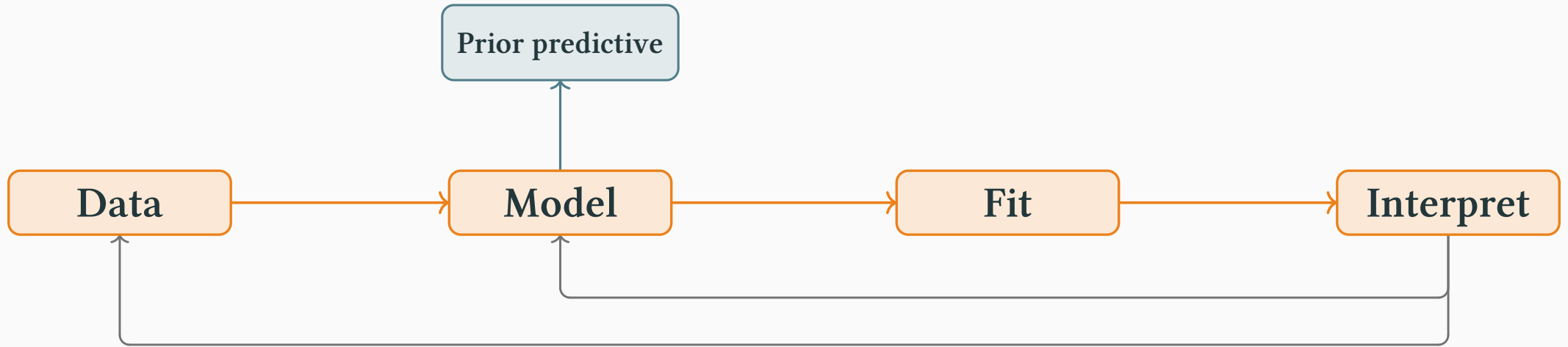
different interpretations · iterative experimental design

Checks make each transition accountable



Validation belongs at each step

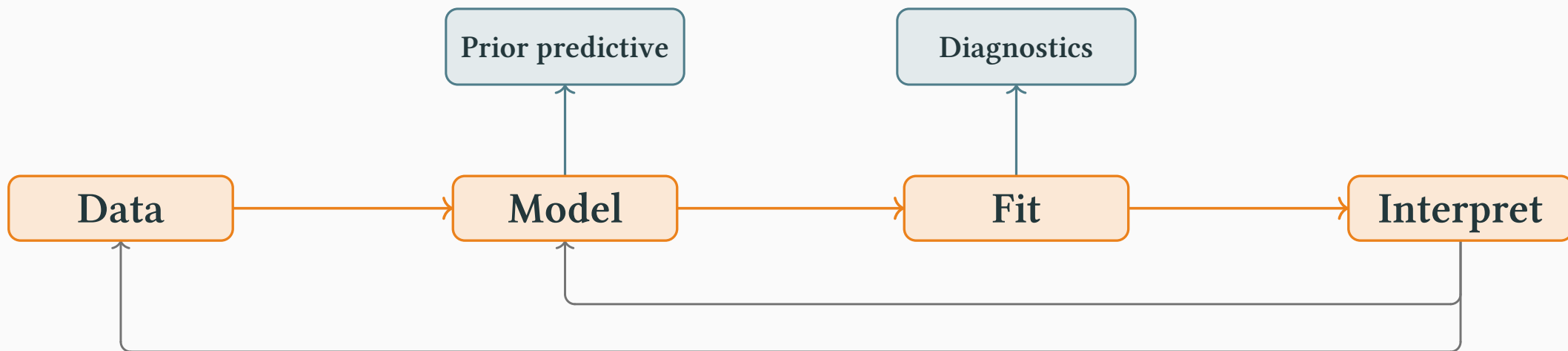
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Prior predictive

tests model coherence

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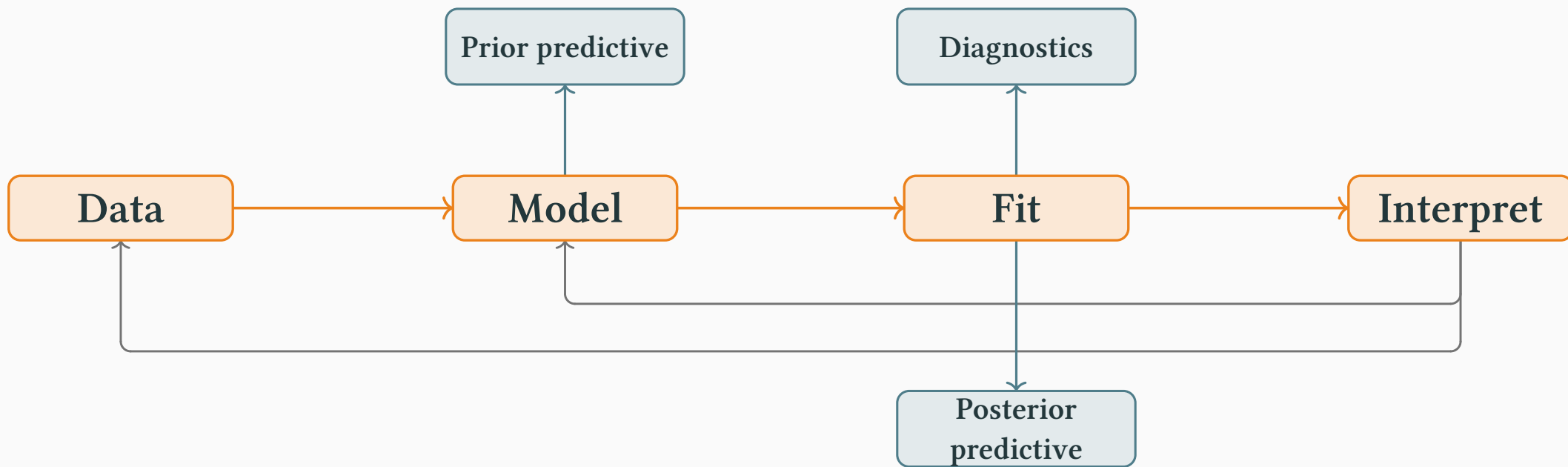
Prior predictive

Diagnostics

tests model coherence

test whether fitting converged

Checks make each transition accountable



Validation belongs at each step

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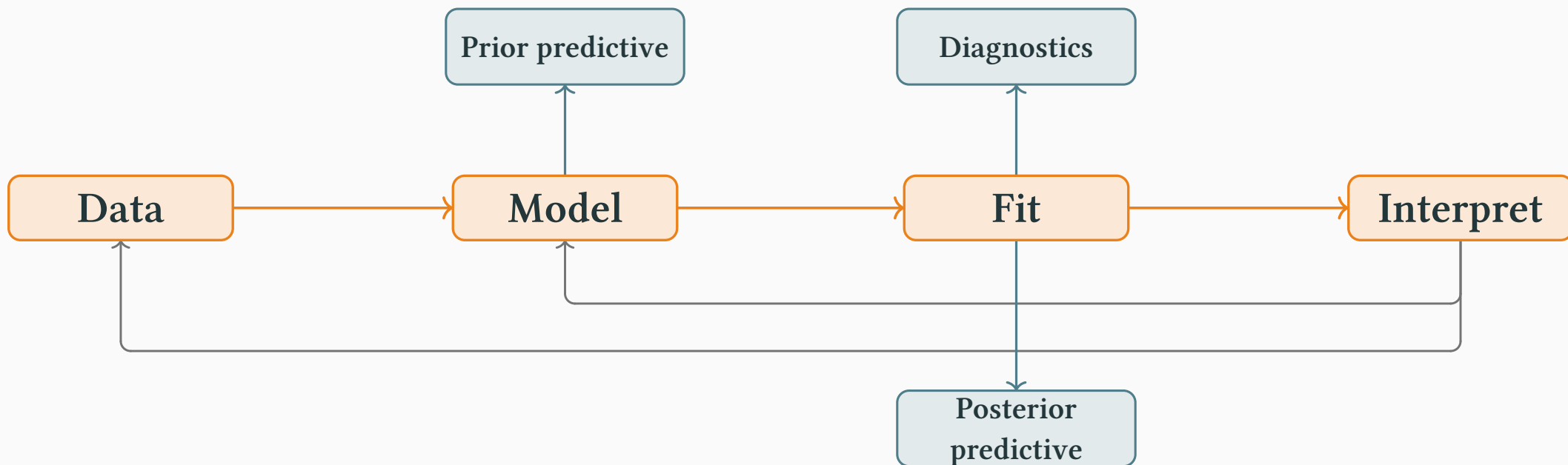
Posterior predictive

tests model coherence

test whether fitting converged

tests fit to the observed data

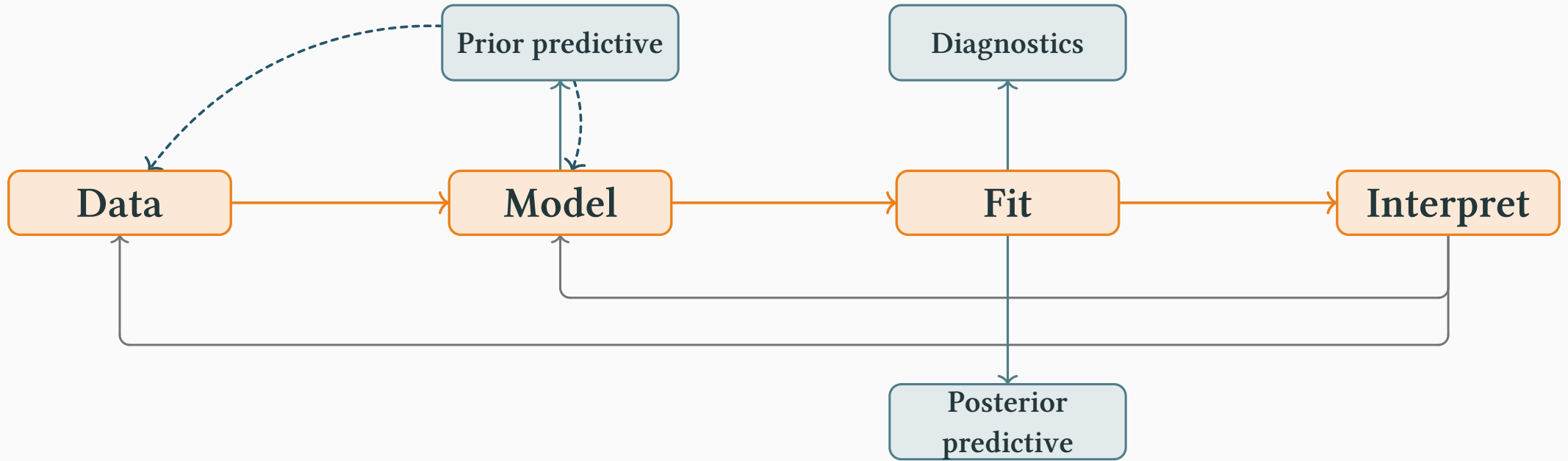
A failed check tells us where to return



A failed check

means something must change

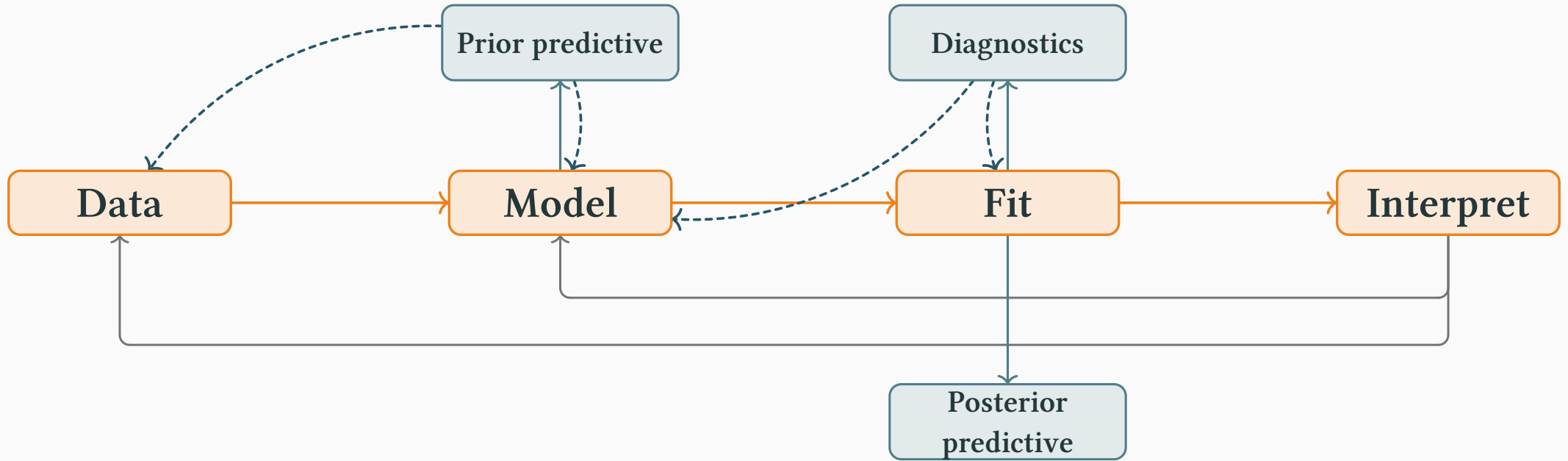
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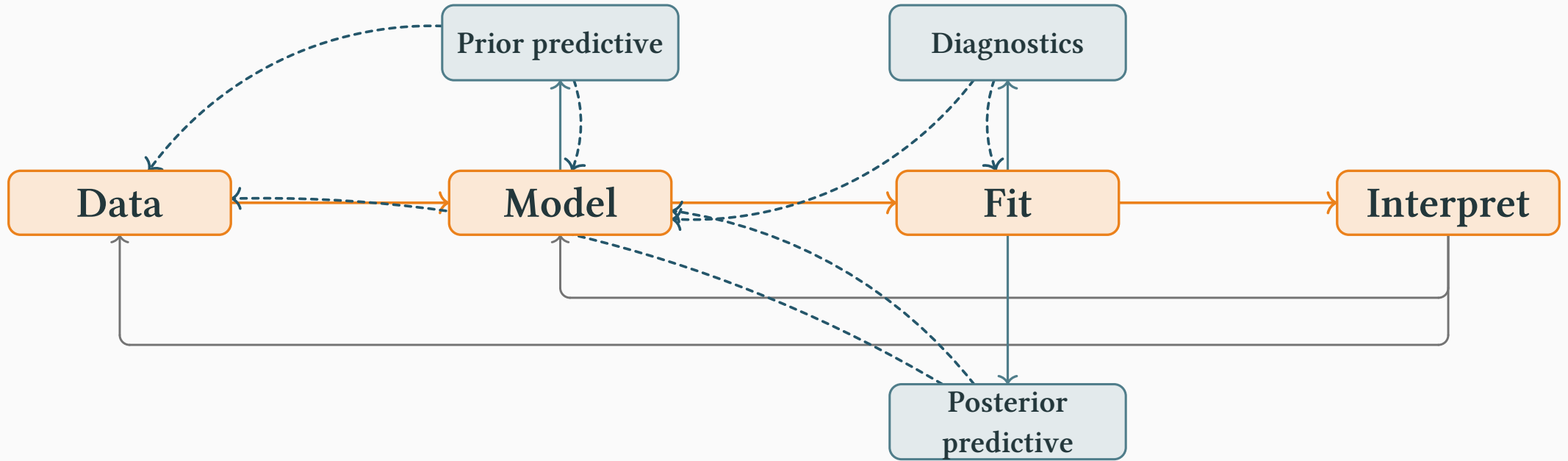
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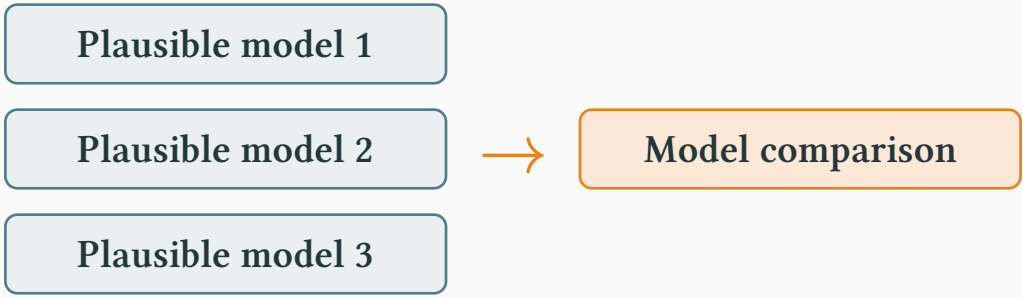
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A failed check
Prior predictive
Diagnostics
Posterior predictive

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revisit assumptions about the model or data
revisit the model or the computation
reassess how the model represents the data

Model comparison asks which useful story predicts best



Leave-one-out cross-validation			
Model	LOO	SE	Δ LOO
Model 1	-124.8	4.5	13.0
Model 2	-112.4	0.3	0.6
Model 3	-111.8	—	0.0

illustrative values; larger LOO is better

The workflow can lead to several **plausible models** with different interpretations. It is also a useful way to think about frequentist “null models.”

Model comparison is an essential tool in Bayesian data analysis.
All models are wrong; some models are informative.

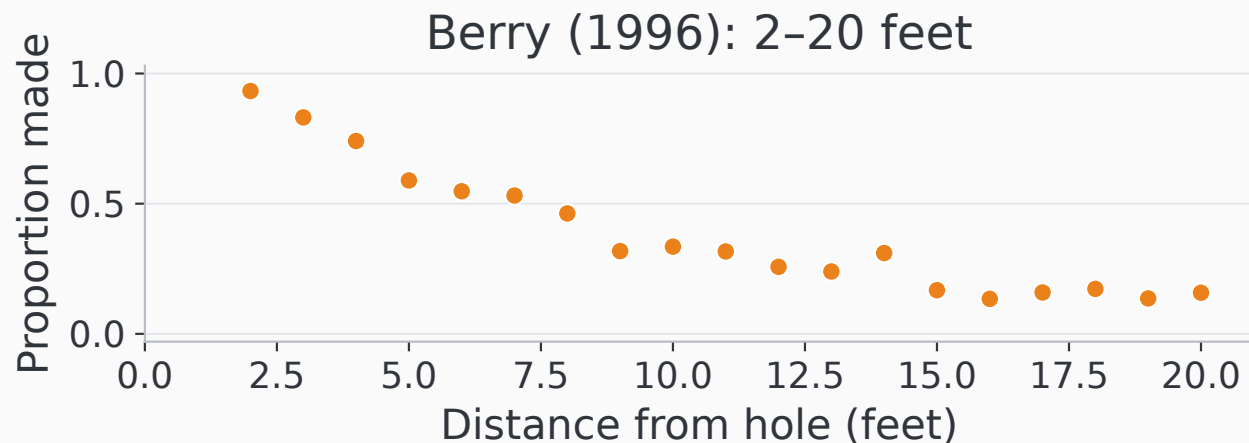
Working through the workflow: golf putting

Two putting datasets let us fit, then genuinely predict



AI-generated course illustration

Berry (1996): professional putting summaries at rounded distances from 2 to 20 feet.



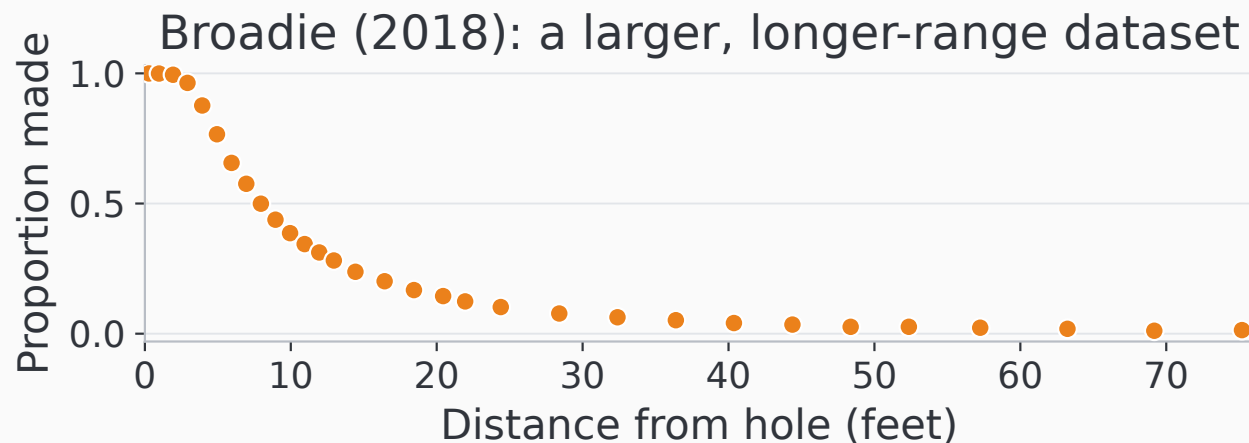
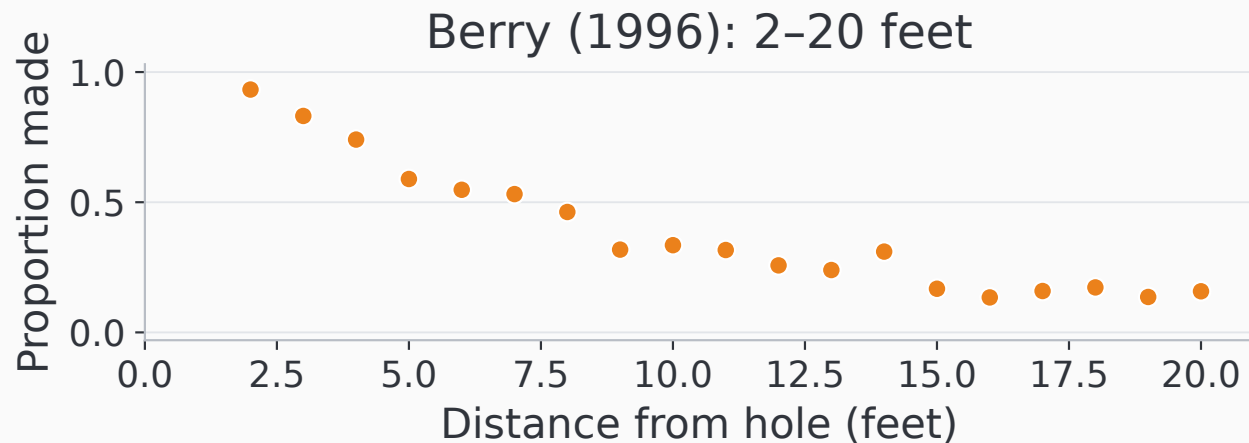
Two putting datasets let us fit, then genuinely predict



AI-generated course illustration

Berry (1996): professional putting summaries at rounded distances from 2 to 20 feet.

Broadie (2018): a later, much larger dataset extending to 75 feet.



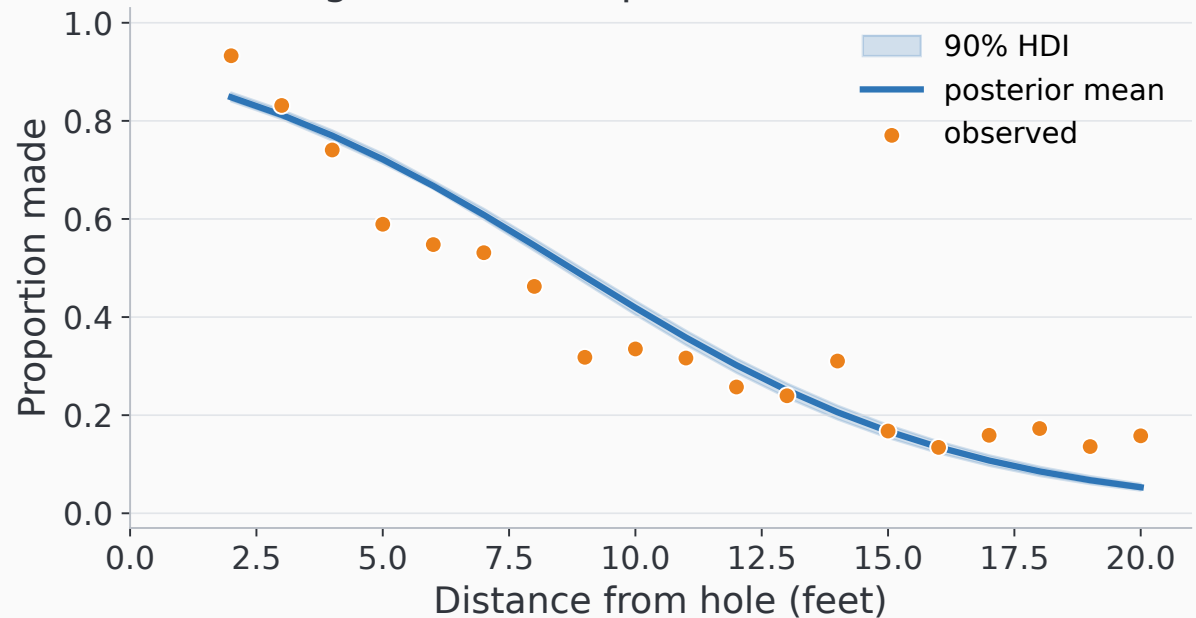
Start simple: logistic regression shows what remains unexplained

$$y_j \sim \text{Binomial}(n_j, p_j)$$

$$\text{logit}(p_j) = \alpha + \beta x_j$$

```
with pm.Model() as model:
    alpha = pm.Normal("alpha", 0, 3)
    beta = pm.Normal("beta", 0, 0.5)
    logit_p = alpha + beta *
distance
    pm.Binomial(
        "made", n=attempts,
        logit_p=logit_p,
        observed=made,
    )
```

The logistic curve captures the broad decline



Start simple. It is usually best to begin with the smallest useful model.

Prior and posterior predictive checks develop intuition for what it can generate.

What it does and does not capture tells us how much remains to explain.

Geometry turns distance into a success probability



Redrawn from Gelman et al., *Bayesian Workflow*, Fig. 25.3

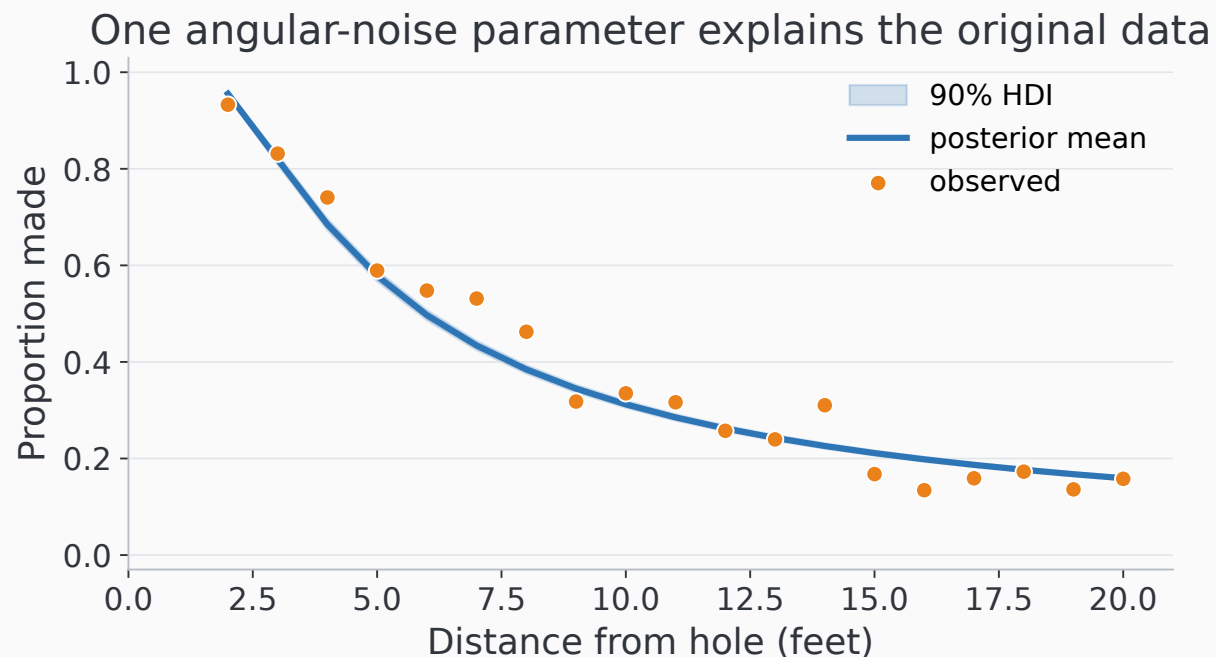
$$\theta(x) = \arcsin\left(\frac{R-r}{x}\right)$$

$$p(x) = 2\Phi\left(\frac{\theta(x)}{\sigma}\right) - 1$$

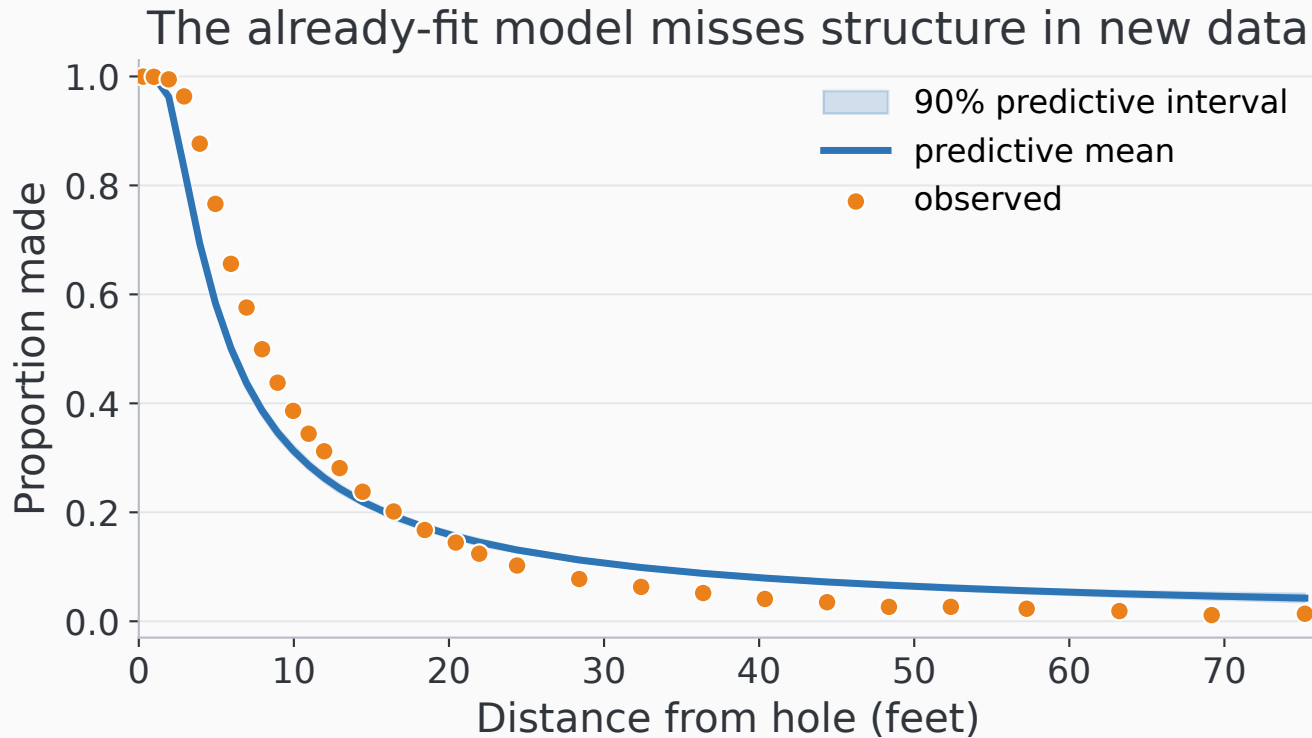
- A highly simplified model of putting physics
- The only parameter is noise in the aiming angle, σ
- These counts overwhelm almost any reasonable prior
- It is still healthy to ask what a reasonable prior implies

One angular-noise parameter fits the original data remarkably well

```
with pm.Model() as model:
    sigma_deg = pm.LogNormal(
        "sigma_deg", log(2), 0.7
    )
    theta = pm.math.arcsin(
        (cup_radius - ball_radius)
        / distance
    )
    p = 2 * pm.math.invprobit(
        theta / deg2rad(sigma_deg)
    ) - 1
    pm.Binomial(
        "made", n=attempts,
        p=p, observed=made,
    )
```



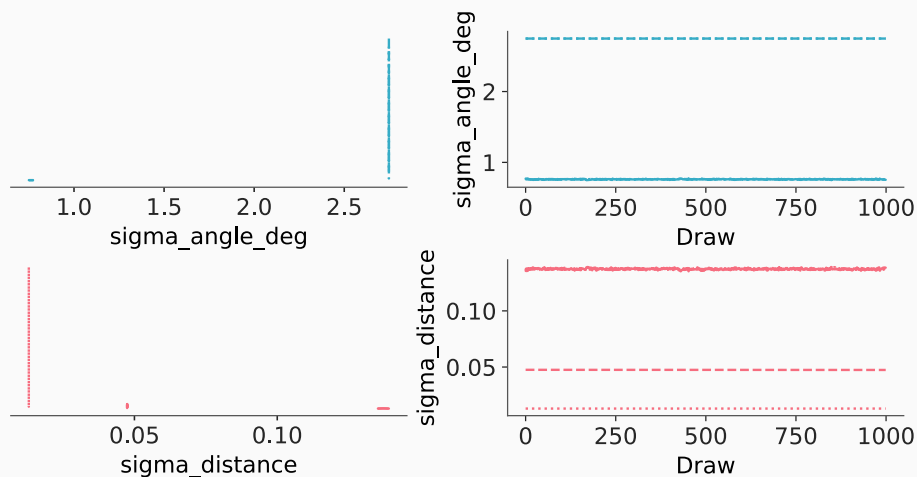
A trusted model earns a harder test on new data



When we have faith in a model, we can test it against new data as those data become available.

Data: Broadie (2018); comparison follows *Bayesian Workflow*, §25.3

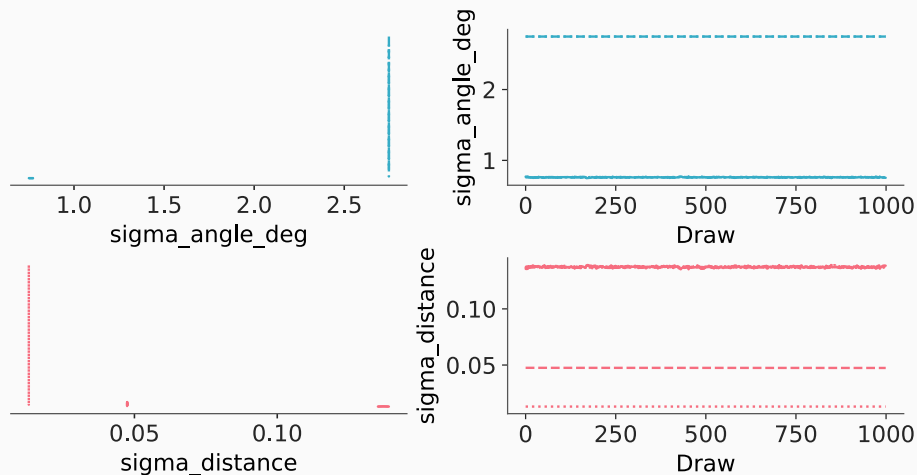
The aiming + distance model: have we sampled the posterior?



The chains disagree:

- $\hat{R} \approx 2$
- ESS ≈ 5 of 4,000 draws

The aiming + distance model: have we sampled the posterior?

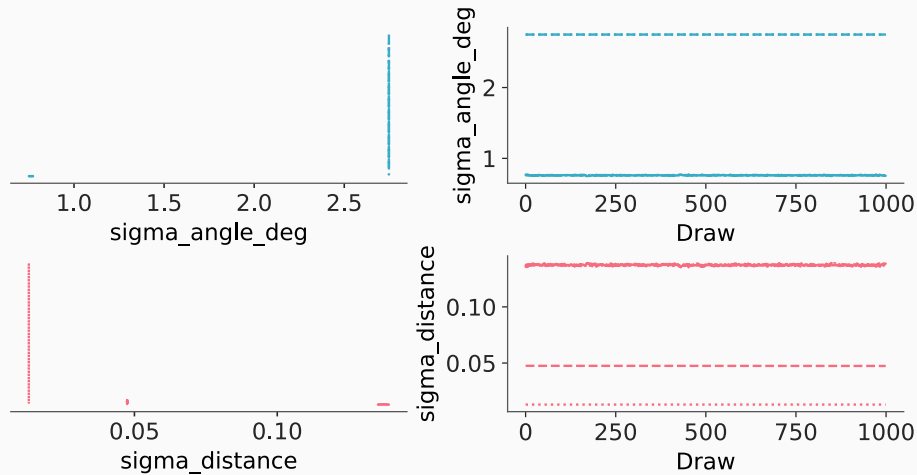


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We have not sampled the posterior.

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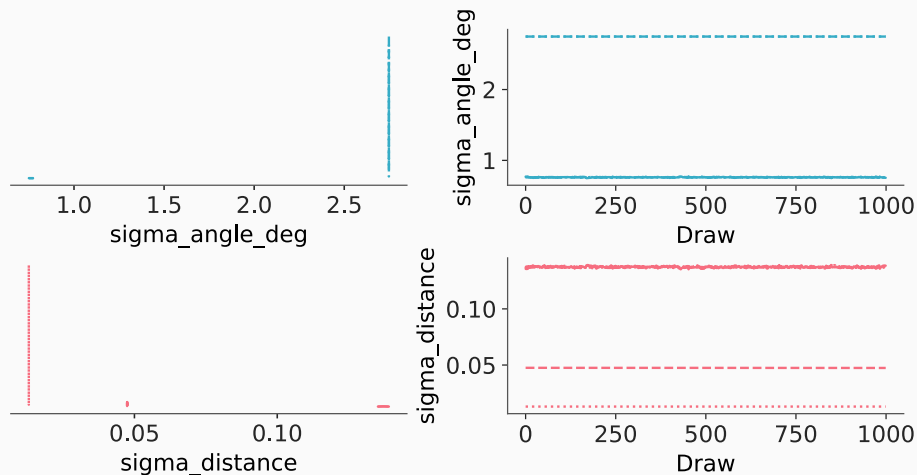
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So model fit is not even the question yet.

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So model fit is not even the question yet.

A good result: the diagnostics caught it.

Continued improvements

- Model misfit results from very large distances in logit scale near the hole
 - Replace the logistic model with a normal model
 - Wrong but better
- Try to return to logit model by adding a distance-dependent variance term
 - Much worse fit
 - Sometimes a ‘truer’ model is not better
- Recover logit model by using proportional rather than additive noise
 - Then use the new model to discover the true distance parameter
- Iterative model building should be central to data analysis
 - Develop intuition
 - Develop critical approach

Becoming independent Bayesian analysts

What we will practice

- Choosing priors
 - Using common sense
 - Using prior predictive checks
 - Practical issues
- Building hierarchical models
 - Separating uncertainty about individuals from uncertainty about the population
 - Understanding how hierarchical models share information across individual fits
 - Choosing priors for population- and individual-level parameters
- Using AI to support Bayesian data analysis